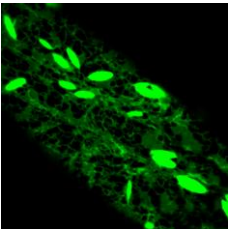


Modality

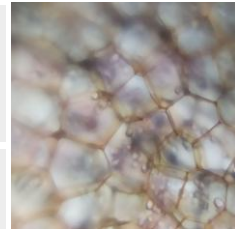


Question: What is the most likely microscopy modality used to acquire this image?

Options: light, fluorescence, electron, none of above

Answer: fluorescence microscopy

Domain

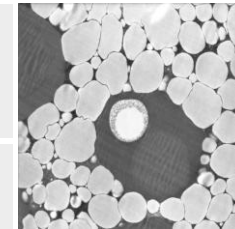


Question: What is the biological domain of this sample?

Options: botany, cell biology, mycology, nematology

Answer: botany

Resin

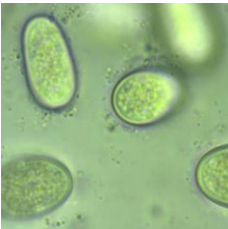


Question: What is the most likely resin used to acquire this image?

Options: spurr, quetol-651, epon, none of above

Answer: epon

Pathogen

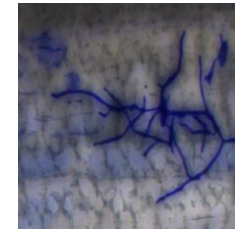


Question: What pathogen does this microscopic image belong to?

Options: brasiliensis, crassispora, exigua, gilanensis

Answer: exigua

Host

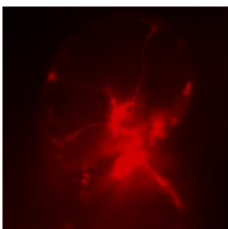


Question: What species does this microscopic image belong to?

Options: cucumber, tomato, barley, rice

Answer: barley

Stain



Question: What is the most likely stain used to acquire this image?

Options: GFP, RFP, Quinacrine, DAPI

Answer: RFP

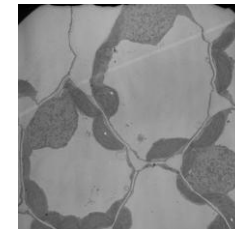


PlantMicro

MICROSCOPIC PLANT VLM BENCHMARK

10 Benchmark tasks
5,210 Images
9,718 VQA pairs

Organ

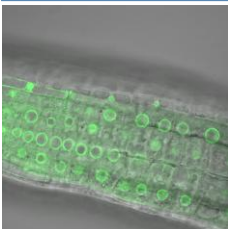


Question: What is the most likely organ this cell is from?

Options: leaf, root, cotyledon, hypocotyl

Answer: leaf

Organelle

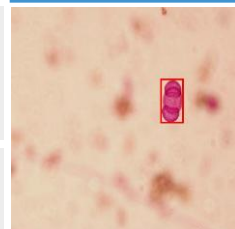


Question: What organelle is in this image?

Options: peroxisome, cytosol, nucleus, plastid

Answer: nucleus

Localization



Question: Please detect the pollen in this image and provide the bounding box coordinates.

Answer: [x1:274, y1:132, x2:312, y2:209]

Counting



Question: How many complete pollen grains are present in this image?

Answer: 10